

Lab 1: Implementation of Adders

Aim

To design and simulate Half Adder, Full Adder, and 4-bit Adder circuits using Logisim Evolution and verify their outputs.

Procedure

Part A: Half Adder

1. Open Logisim Evolution software.
2. Place two input pins (A and B).
3. Use an XOR gate to generate the Sum output.
4. Use an AND gate to generate the Carry output.
5. Connect outputs to LEDs.
6. Simulate the circuit for all input combinations and observe outputs.

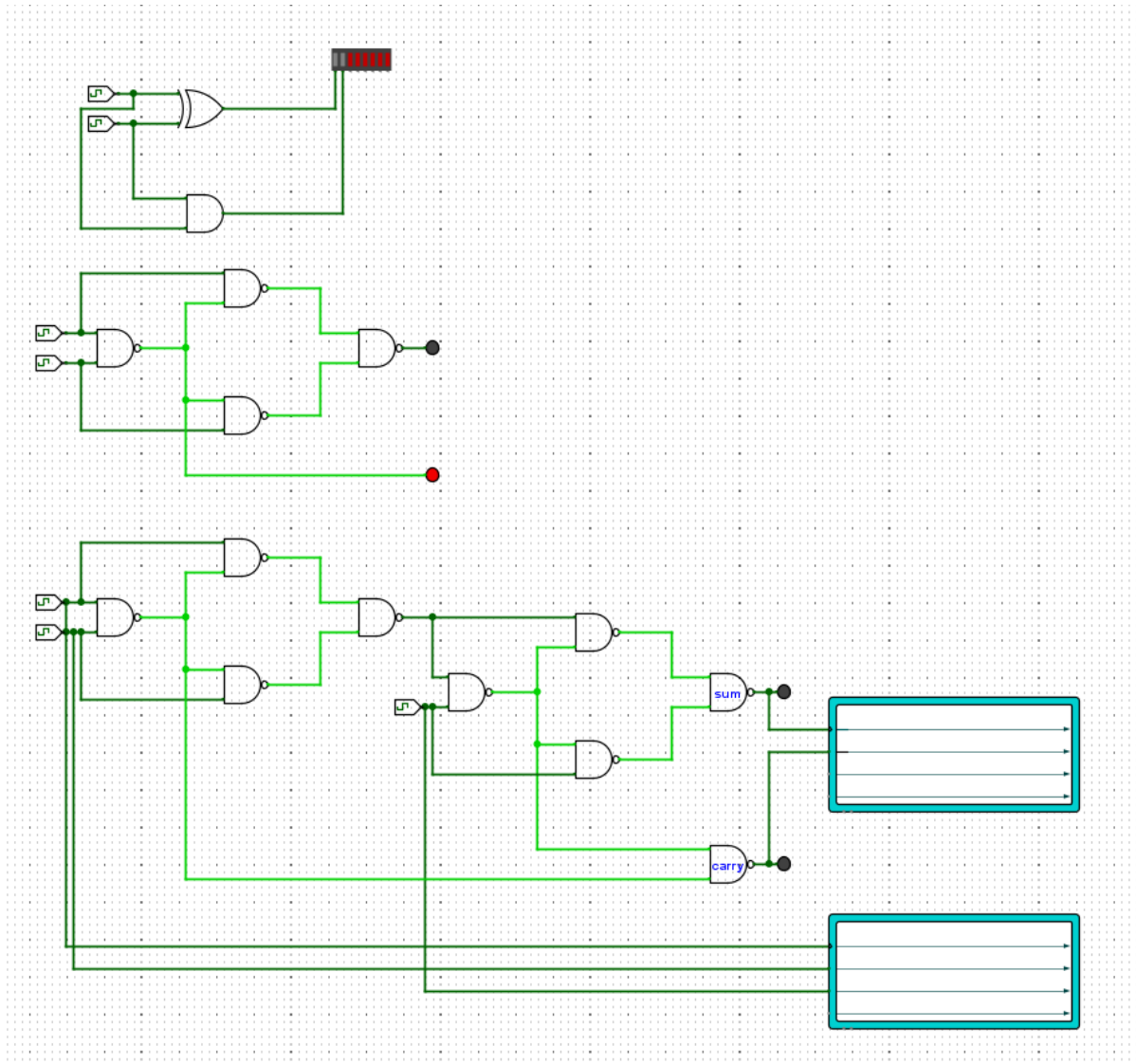
Part B: Full Adder

1. Place three input pins (A, B, Cin).
2. Construct the circuit using two Half Adders and one OR gate:
 - First Half Adder takes inputs A and B.
 - Second Half Adder takes Sum output of first and Cin.
 - OR gate combines carry outputs of both Half Adders.
3. Connect Sum and Carry outputs to LEDs.
4. Simulate for all input combinations and verify results.

Part C: 4-bit Adder

1. Use four Full Adders to build a 4-bit adder.
2. Connect inputs A0–A3 and B0–B3.
3. Connect carry-out of each stage to carry-in of next stage.

4. Take final outputs as Sum (S0–S3) and final Carry-out.
5. Apply different binary inputs and observe the output.



Result

The Half Adder, Full Adder, and 4-bit Adder circuits were successfully designed and simulated using Logisim Evolution.

- The Half Adder correctly produced Sum and Carry for all input combinations.
- The Full Adder verified correct operation for three inputs including carry-in.
- The 4-bit Adder successfully performed binary addition with proper carry propagation.

All outputs matched the expected theoretical results.